

Paper I - Basic Sciences in Urology (MCh Urology, Summer 2015)

Contents

1. I-1. Patho-physiology of urine production and transport (LAQ)
 2. I-2. Female pelvic anatomy in relation to prolapse and SUI (LAQ)
 3. I-3a. Newer generation CT scan
 4. I-3b. History of laparoscopy
 5. I-3c. Renin-angiotensin system
 6. I-3d. Patho-physiology of brain stem death
 7. I-3e. Male reproductive immunology
 8. I-3f. Principles of antimicrobial therapy in UTI
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I-1. Describe in details basic patho-physiology of urine production and transport

Type: LAQ (25 marks)

Introduction

- Urine is formed by filtration, reabsorption, secretion and concentration, then moved by peristalsis to a low-pressure bladder.
- Kidney roles: excretion of waste and drugs; volume, water, electrolyte and acid-base balance.

A. Functional anatomy

- Glomerular barrier (fenestrated endothelium, basement membrane, podocytes) restricts by size and negative charge, so albumin is excluded.
- **Juxtamedullary nephrons** (long loops) build the medullary gradient. Juxtaglomerular apparatus: macula densa, arterioles, renin-secreting cells.

B. Renal blood flow (RBF) and autoregulation

- RBF ~25% of cardiac output. Medullary oxygen tension is only 10–20 mmHg, so the thick ascending limb and S3 segment are hypoxia-prone.
- **Autoregulation** keeps RBF and GFR (glomerular filtration rate) constant at mean arterial pressure 80–180 mmHg:
 - **Myogenic:** the afferent arteriole constricts when stretched and dilates (prostaglandins) when pressure falls; NSAIDs (non-steroidal anti-inflammatory drugs) abolish this.
 - **Tubuloglomerular feedback:** low NaCl at the macula densa releases renin; angiotensin II constricts the efferent arteriole. Angiotensin blockade abolishes this.

C. Glomerular filtration

- GFR ~125 mL/min (180 L/day); filtration fraction ~20%.
- **Net filtration pressure** $\approx$ glomerular hydrostatic 60 – Bowman's space 18 – oncotic 32 = **~10 mmHg**.
- Efferent constriction lowers RBF but **raises** GFR; afferent constriction lowers both. Obstruction raises Bowman's space pressure, so GFR falls.
- **Also know:** GFR estimated from creatinine or cystatin C (race-free 2021 equations).

D. Tubular handling

Segment	Handling	Drug link
Proximal tubule	~ 65% Na ⁺ and water; all glucose (SGLT2, sodium-glucose cotransporter 2); bicarbonate	SGLT2 inhibitors
Thick ascending limb	~ 25% NaCl (NKCC2); water-impermeable "diluting segment"	Loop diuretics
Distal convoluted tubule	5–7% Na ⁺ ; Ca ²⁺	Thiazides
Collecting duct	Principal cells: Na ⁺ /K ⁺ (aldosterone), water; intercalated cells: acid	Spironolactone; distal RTA (renal tubular acidosis)

E. Concentration

- **Countercurrent multiplication:** ascending limbs absorb NaCl without water, raising medullary osmolality from 300 to ~**1200 mOsm/kg**; urea recycling adds about half. Vasa recta exchange without washing out the gradient.
- **Antidiuretic hormone (ADH)**, released by high osmolality or low volume, acts on V2 receptors to insert aquaporin-2 (AQP2) into principal cells. Concentration needs both a hypertonic medulla and AQP2.
- Aldosterone retains Na⁺; atrial natriuretic peptide excretes it.

F. Ureteric transport

- **Pacemaker:** atypical smooth muscle cells at the minor calyces/pelviccalyceal border.
- Peristalsis is **myogenic**: the excised or denervated ureter keeps contracting. Nerves (T11–L1, S2–4) carry pain referred to flank, groin and genitalia.
- Calcium entry triggers action potentials spread by gap junctions; calcium-calmodulin then activates myosin light-chain kinase.
- Walls coapt behind each bolus: 2–6 waves/min at 20–80 cmH₂O. Very high flow makes boluses coalesce, dilating the ureter **without obstruction**.
- Ureterovesical tunnel length:diameter 5:1 prevents reflux; bladder pressure >40 cmH₂O stops efflux.
- Endotoxin inhibits and α1-blockers relax the ureter.

G. Bladder storage and voiding

- **Storage** below ~10 cmH₂O: compliance (collagen, elastin), sympathetic β3 detrusor relaxation, α1 bladder-neck tone, pudendal sphincter tone.
- **Voiding:** pelvic parasympathetic (S2–4) acetylcholine on M3 receptors contracts the detrusor; nitric oxide relaxes the urethra. Aδ afferents sense fullness.

H. Pathophysiology of obstruction (unilateral: triphasic)

Phase 1 (0–1.5 h): RBF up, pressure up (afferent dilatation)
Phase 2 (1.5–5 h): RBF down, pressure up (efferent constriction)
Phase 3 (>5 h): RBF down, pressure down (afferent constriction)

- Ischaemic injury starts **within 2 hours** of complete obstruction.
- **Tubules:** down-regulated aquaporins and Na⁺ transporters cause a concentrating defect, salt wasting and distal RTA; K⁺ is wasted early, retained late.
- **Fibrosis** (final path to failure): stretch, inflammation, TGF-β (transforming growth factor-β), angiotensin II, microRNAs.
- Immature kidneys fail to develop; mature kidneys proliferate, then scar.

- **Clinical:** hypertension (~77% bilateral, ~20% unilateral); contralateral hypertrophy; post-obstructive diuresis ~2% unilateral vs ~50% bilateral.
- **Lower tract:** detrusor hypertrophy, then decompensation; collagen lowers compliance.
- **Also know:** diuretic MAG3 (mercaptoacetyl triglycine) renogram (half-time >40 min = obstructed); Whitaker test (>15–20 cmH₂O = obstructed); SGLT2 inhibitors and tolvaptan as recent renal drugs.

Diagram to draw: Nephron with transporters; triphasic graph.

Key points to remember:

- RBF 25% of cardiac output; GFR 125 mL/min; net filtration pressure ~10 mmHg.
- Proximal 65%, loop 25%, distal 5–7% of Na⁺; ADH → V2 → AQP2.
- Myogenic peristalsis from calyceal pacemaker; post-obstructive diuresis 2% vs 50%.

I-2. Describe female pelvic anatomy in relation to prolapse and SUI

Type: LAQ (25 marks)

Introduction

- Support comes from the **levator ani, endopelvic fascia and ligaments, perineal membrane and bony pelvis**. Failure causes **pelvic organ prolapse (POP)** and **stress urinary incontinence (SUI)**, effort leakage from failed urethral support or closure.

A. Bony pelvis and perineum

- Fixation points and dangers: **sacrospinous ligament** (pudendal bundle), **Cooper's ligament** (bleeding), **sacral promontory** (left common iliac vein, right ureter).
- Perineal membrane** and **perineal body** give distal support.

B. Pelvic floor muscles

Muscle	Attachment	Function
Pubococcygeus	Pubis to vagina, perineal body, anus (coccygeus is separate)	Closes hiatus, lifts urethra
Puborectalis	Sling behind anorectal junction	Anorectal angle
Iliococcygeus	Arcus tendineus levator ani to coccyx	Horizontal levator plate

- Levator tone closes the **urogenital hiatus**, through which all POP passes.
- Muscle fails first, fascia second**: once injury opens the hiatus, ligaments carry the load and fail.
- Up to **20% of primiparous women** have levator defects on imaging; avulsion predicts advanced prolapse.

C. Endopelvic fascia and ligaments

- Cardinal and uterosacral ligaments** suspend the apex; the uterosacral is the main anchor (ureter close by at the cervix).
- Pubocervical fascia attaches laterally to the **arcus tendineus fasciae pelvis (ATFP, "white line")**, pubis to just above the ischial spine.
- Rectovaginal fascia** supports the posterior wall; **pubourethral ligaments** anchor the mid-urethra.

D. DeLancey's levels of support (1992)

Level	Region	Structures	Defect causes
I - suspension	Cervix/upper third	Cardinal-uterosacral complex	Uterine/vault prolapse, enterocele
II - attachment	Middle third	ATFP, levator fascia	Cystocele, rectocele
III - fusion	Distal third	Perineal membrane, perineal body	Urethral hypermobility, perineal descent

- Central cystocele**: midline weakness, rugae lost. **Paravaginal cystocele**: ATFP detachment (commonest at the ischial spine, Richardson 1981), rugae preserved.
- Level I is the keystone: apical loss underlies most anterior prolapse, so correct the apex in any repair.

E. Urethra and continence

- Continence zone is the **proximal and mid-urethra** (bladder neck weak): striated sphincter (with compressor urethrae and urethrovaginal sphincter) plus oestrogen-sensitive submucosa (~30% of closure).

Theory	Mechanism	Implication
Pressure transmission (Enhörning 1961)	Hypermobility urethra leaves the abdominal pressure zone	Restore position (Burch)
Hammock (DeLancey 1994)	Urethra compressed against firm vaginal backing	Restore backing
Integral theory (Petros & Ulmsten 1990)	Pubourethral ligament, suburethral hammock, pubococcygeus at mid-urethra	Mid-urethral sling
Intrinsic sphincter deficiency (ISD) (McGuire 1978)	Urethral function fails	Sling, bulking

- ISD: Valsalva leak point pressure <60 or maximal urethral closure pressure <20 cmH₂O. Most SUI has some ISD, so slings work in hypermobility but Burch does poorly in severe ISD.

F. Innervation

- Pudendal nerve (S2-4):** sphincters; near the ischial spine. **Nerve to levator ani (S3-5):** on the muscle's pelvic surface; neuropathy in about a quarter of primiparous women.

G. Risk factors (Bump & Norton)

- Predisposing: genetics, collagen disorders. Inciting: **vaginal delivery** (strongest; forceps worse), hysterectomy. Promoting: obesity, constipation, cough. Decompensating: ageing.

H. Staging and examination

POP-Q (Pelvic Organ Prolapse Quantification, Bump 1996) stage	Most distal point
0	No prolapse
I	>1 cm above hymen
II	Within ± 1 cm of hymen
III	>1 cm below hymen, < (total vaginal length - 2) cm
IV	Complete eversion

- Also know:** Baden-Walker grades 0-4; stage I-II often normal (POSST study); cough stress test with prolapse reduced (occult SUI); Q-tip >30° = hypermobility.

I. How anatomy guides surgery

Defect	Operation	Principle
SUI	Mid-urethral sling: retropubic (Ulmsten 1995, TVT [tension-free vaginal tape]) or transobturator (Delorme 2001)	Mid-urethral support
SUI	Burch colposuspension	Fascia to Cooper's ligament
SUI - ISD	Autologous sling at bladder neck, bulking	Compression
Paravaginal cystocele	Paravaginal repair	Reattach to ATFP
Apical	Sacrocolpopexy (most durable)	Mesh to ligament over S1

Defect	Operation	Principle
Apical (vaginal)	Sacrospinous fixation 1.5–3 cm medial to spine	Avoids pudendal bundle

- **April 2019:** the US FDA (Food and Drug Administration) stopped transvaginal mesh for anterior POP; slings unaffected.

Diagram to draw: DeLancey levels I–III; hammock cross-section.

Key points to remember:

- DeLancey: I suspension, II attachment (ATFP), III fusion; level I is the keystone.
- Enhorning 1961, DeLancey 1994, Petros & Ulmsten 1990, McGuire ISD 1978.
- Central cystocele rugae lost, paravaginal preserved; POP-Q II = ± 1 cm of hymen.

I-3a. Newer generation CT scan

Type: SAQ (10 marks)

Introduction

- Computed tomography (CT) is the reference imaging for stones, masses and trauma. Attenuation is expressed in Hounsfield units (HU): water 0, air –1000.

Evolution

Generation	Feature
Helical (~1989)	Continuous table movement, volumetric data
Multidetector (1998 → 320-row)	Sub-millimetre slices, 3D reconstruction
Dual-source/dual-energy (2005)	Two energies: material analysis
Photon-counting (US approval 2021)	Energy-sorts each photon: less noise, dose and metal artefact

Dual-energy CT (DECT)

- Two energies (e.g. 80 and 140 kVp) compared pixel by pixel: **uric acid vs non-uric-acid stones, virtual unenhanced** images (one phase fewer) and **iodine maps** (hyperdense cyst vs enhancing mass).

Low-dose CT

Protocol	Dose
Conventional abdomen/pelvis	10–20 mSv
Low-dose	<3–4 mSv
Ultra-low-dose	1–2 mSv

- Stone sensitivity stays ~95%; use standard dose if body mass index >30. EAU (European Association of Urology): ultrasound first; AUA (American Urological Association): CT first.

Reconstruction

- Filtered back projection is noisy; iterative reconstruction halves dose; **deep learning reconstruction** gives good images at ~1 mSv.

Protocols

- CT urography:** non-contrast, nephrographic and excretory phases (~20–30 mSv). **Split-bolus** merges phases, lowering dose but may miss subtle urothelial lesions.
- Renal mass:** enhancement ≥20 HU = solid, 10–20 equivocal; nephrographic phase most sensitive; Bosniak 2019 for cysts.
- Also know:** stone HU (uric acid 200–400, calcium oxalate 600–1200; indinavir invisible); bone window for fragments; CT angiography for donors; adrenal washout protocol; partial volume causes pseudoenhancement of small cysts; ALARA (as low as reasonably achievable) dosing, since radiation risk has no threshold.

Diagram to draw: Timeline: axial → helical → multidetector → dual-source → photon-counting.

Key points to remember:

- Doses: conventional 10–20, low-dose <3–4, ultra-low 1–2 mSv.
- DECT: uric acid stones, virtual unenhanced, iodine maps.
- Enhancement ≥20 HU = solid; deep learning enables ~1 mSv CT.

I-3b. History of Laparoscopy

Type: SAQ (10 marks)

Introduction

- Laparoscopy is endoscopic viewing of the peritoneal cavity through a pneumoperitoneum; it grew from optics to robotic surgery.

Optics and access

Year	Pioneer	Milestone
1806	Bozzini	Lichtleiter , candle-lit light conductor
1877	Nitze	Distal electric light and lenses
1901	Kelling	Insufflated abdomen of a dog
1910	Jacobaeus	First human series; "laparoscopy"
1938	Veress	Spring-loaded needle
1966	Hopkins	Rod-lens system
1970s-80s	Semm	Automatic insufflator; operative laparoscopy
1970s	Hasson	Open access
1987	Mouret	Video cholecystectomy

- Also know:** Desormeaux 1853, first therapeutic endoscopy; fibreoptics (Tyndall's principle) for flexible scopes; CO₂ became the standard insufflant.

Urological milestones

Year	Pioneer	Milestone
1976	Cortesi	Impalpable testis, first urological use
1990-91	Clayman	First laparoscopic nephrectomy ; nephroureterectomy 1991
1992	Gaur (India)	Balloon retroperitoneal access
1995	Ratner	Live-donor nephrectomy
1997	Schuessler	Laparoscopic radical prostatectomy (slow, not adopted)
2000	Guillonneau & Vallancien	Standardised prostatectomy
2000	da Vinci; Binder	Robot cleared in US; first robotic prostatectomy
2000s	Menon	Vattikuti Institute Prostatectomy
2007	Raman; Rane	Single-port nephrectomy

Why the robot spread

- Laparoscopic prostatectomy stalled because intracorporeal suturing was hard. The robot gave wristed instruments and a 3D magnified view.
- da Vinci generations: S 2006 (fourth arm), Si 2009 (dual console), Xi 2014 (flexible docking), SP 2018 (single port).

Indian contribution and recent advances

- Gaur's balloon made retroperitoneoscopy practical; indigenous robot SSI Mantra with telesurgery.
- New robots (Versius, Hugo, Hinotori), telesurgery, haptic feedback.

Diagram to draw: Timeline 1806 → 1901 → 1910 → 1938 → 1990 → 2000.

Key points to remember:

- Bozzini 1806; Kelling 1901 (dog); Jacobaeus 1910 (human); Veress 1938; Hopkins rod-lens.
 - Urology: Cortesi 1976; Clayman 1990; Gaur 1992; Ratner 1995.
 - da Vinci 2000; Binder first robotic prostatectomy; robot solved suturing barrier.
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I-3c. Renin-angiotensin system

Type: SAQ (10 marks)

Definition

- Hormonal cascade controlling **blood pressure, extracellular volume, Na⁺/K⁺ balance and GFR (glomerular filtration rate)**.

Cascade

Angiotensinogen (liver)
↓ renin (juxtaglomerular cells) – rate-limiting
Angiotensin I
↓ ACE (angiotensin-converting enzyme), mainly lung
Angiotensin II → AT1 / AT2 receptors
↓ ACE2
Angiotensin (1-7) → Mas receptor (protective)

Juxtaglomerular apparatus and renin release

- Granular cells in the afferent arteriole secrete renin and act as baroreceptors; the **macula densa** senses tubular NaCl.
- **Stimuli:** low perfusion pressure, low macula densa NaCl, sympathetic β_1 activity.
Inhibitors: angiotensin II feedback, atrial natriuretic peptide, β -blockers.

Actions of angiotensin II

- **AT1 receptor:** vasoconstriction; **aldosterone** (distal Na⁺ retention, K⁺ loss); direct proximal Na⁺ reabsorption; thirst and ADH (antidiuretic hormone); **fibrosis**.
- **AT2 receptor:** vasodilatation; fetal urinary tract development.
- **Intrarenal:** efferent constriction maintains GFR when perfusion falls, so ACE inhibitors and ARBs (angiotensin receptor blockers) drop GFR in bilateral or solitary-kidney renal artery stenosis.

Urological relevance

- **Renovascular hypertension (Goldblatt kidney):** atherosclerosis or fibromuscular dysplasia; **renal vein renin ratio >1.5** lateralises.
- **Page kidney:** subcapsular haematoma compresses parenchyma; ACE inhibitor/ARB, then drainage.
- **Reninoma (juxtaglomerular cell tumour):** young women, ≤ 3 cm mass, refractory hypertension, **hypokalaemia**; excision cures most.
- **Obstruction:** hypertension in $\sim 77\%$ bilateral vs $\sim 20\%$ unilateral; angiotensin II drives fibrosis.
- **Also know:** post-traumatic hypertension (arteriovenous fistula, scarring); reflux nephropathy (renin-guided segmental resection); angiotensin blockade and hyperkalaemia in CKD (chronic kidney disease); ACE inhibitors/ARBs contraindicated in pregnancy.

Diagram to draw: Juxtaglomerular apparatus and cascade.

Recent advances

- Renal denervation (US approval 2023) for resistant hypertension.

Key points to remember:

- Renin is rate-limiting; released by low pressure, low NaCl, $\beta 1$.
 - Efferent constriction maintains GFR, so ACE inhibitors risk renal failure in bilateral stenosis.
 - Renin ratio >1.5 lateralises; reninoma excision cures most.
-

I-3d. Patho-physiology of Brain stem death

Type: SAQ (10 marks)

Definition

- **Irreversible loss of the capacity for consciousness and to breathe** from permanent brainstem failure; legally death. Donation after brain death (DBD) keeps circulation until procurement, minimising warm ischaemia.

Pathophysiology

Brain injury → oedema → raised intracranial pressure (ICP)
→ falling cerebral perfusion → herniation ("coning")
→ Pons: Cushing reflex (hypertension, bradycardia)
→ Medulla: catecholamine storm (hypertension, arrhythmia)
→ Loss of sympathetic tone → vasodilatation, hypotension

- The storm injures myocardium and causes pulmonary oedema. Loss of antidiuretic hormone causes **diabetes insipidus (DI)**: polyuria, hypernatraemia; thyroid hormone and cortisol fall.
- **Hypothermia** (hypothalamic loss) and **disseminated intravascular coagulation (DIC)** from brain thromboplastin.

Diagnosis

- **Preconditions:** apnoeic coma of known irreversible cause; exclude sedatives, relaxants, hypothermia (35–36 °C by protocol), metabolic causes.
- **Brainstem reflexes, all must be absent:**

Test	How it is done	Cranial nerves
Pupillary light reflex	Pupils fixed, no response to bright light	II, III
Corneal reflex	Touch cornea; no blink	V, VII
Oculovestibular (caloric)	~50 mL ice-cold water into each ear (drum visible); no eye movement	VIII, III, VI
Oculocephalic (doll's eye)	Turn head; eyes move with the head	VIII, III, VI
Motor response	Central painful stimulus (supraorbital pressure); no facial grimace	V, VII
Gag and cough	Stimulate pharynx; pass suction catheter down trachea; no response	IX, X

- **Apnoea test:** pre-oxygenate with 100% oxygen, disconnect the ventilator with oxygen insufflation, and watch for breathing effort until **arterial CO₂ (PaCO₂) ≥60 mmHg**. No effort = positive. Abort if hypotension, desaturation or arrhythmia.
- **Ancillary tests** (electroencephalography (EEG), cerebral angiography, perfusion scan, transcranial Doppler) are not mandatory; used when clinical testing cannot be completed.

Indian law

- **THOA (Transplantation of Human Organs Act) 1994**, amended 2011, Rules 2014: **4-member board** (hospital in-charge, independent specialist, neurologist/neurosurgeon, treating doctor; no transplant team member); **two tests at least 6 hours apart**.

Donor management

- **Rule of 100s:** systolic pressure >100 mmHg, urine >100 mL/h, PaO₂ (arterial oxygen tension) >100 mmHg, haemoglobin >100 g/L.
- **Vasopressin** preferred (also treats DI); thyroid hormone, methylprednisolone, insulin.
- **Also know:** cold storage at 4 °C (University of Wisconsin solution); donor exclusions (metastatic cancer, melanoma, untreatable viral infection).

Recent advances

- Hypothermic machine perfusion reduces delayed graft function.

Diagram to draw: Herniation sequence leading to storm, DI, hypothermia, DIC.

Key points to remember:

- Raised ICP → Cushing reflex → storm → hypotension, DI, hypothermia, DIC.
 - Apnoea test: no breathing at PaCO₂ ≥60 mmHg.
 - THOA: 4-member board, two tests ≥6 h apart; Rule of 100s.
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I-3e. Male reproductive immunology

Type: SAQ (10 marks)

Introduction

- Haploid germ cells appear at puberty, after self-tolerance is established, so their antigens are "foreign". The testis is therefore **immune-privileged**.

Immune privilege

Anatomical barrier: blood-testis barrier (BTB)

- **Tight junctions between Sertoli cells**, formed at puberty, divide a **basal compartment** (spermatogonia, plasma-like fluid) from an **adluminal compartment** (later spermatocytes, spermatids).

Functional barrier

- A small antigen leak at the **rete testis, efferent ductules and epididymis** induces suppressor T cells and tolerance; a large load provokes immunity.
- Sertoli and local mediators: galectin-1, TGF- β (transforming growth factor- β), interferon- γ , soluble Fc receptor, **androgens**.
- Insult before puberty does not induce antibodies; the barrier also limits chemotherapy (isolated testicular relapse).

Antisperm antibodies (ASA)

- **Causes:** obstruction (**vasectomy** commonest), torsion, trauma, biopsy, infection, heat (varicocele, cryptorchidism), genetic (human leukocyte antigen HLA-B28).
- **Effects:** agglutination, poor motility, failure to cross cervical mucus and bind the zona; head-bound antibody matters most.
- **Tests:** mixed antiglobulin reaction or immunobead test; **$\geq 50\%$ of motile sperm bound** is significant (WHO [World Health Organization] 2010).
- **Treatment:** corticosteroid trials conflict; washing removes only unbound antibody; **IVF/ICSI (in vitro fertilisation/intracytoplasmic sperm injection)** is usual.

Infection and inflammation

- **Leukocytospermia:** $>1 \times 10^6$ peroxidase-positive leukocytes/mL. Reactive oxygen species and cytokines damage sperm.
- Organisms: gonococcus, *Chlamydia*, *E. coli*; mumps orchitis after puberty; tuberculosis scars ducts.
- **Also know:** sperm DNA fragmentation testing; prostatitis harms sperm via oxidative stress and ASA; epididymal macrophages ingest sperm after vasectomy; seminal plasma suppresses female tract immunity and is antioxidant.

Diagram to draw: Seminiferous tubule with Sertoli tight junctions and compartments.

Key points to remember:

- BTB = Sertoli tight junctions; plus active tolerance (galectin-1, TGF- β , androgens).
- ASA: vasectomy commonest; $\geq 50\%$ bound significant; IVF/ICSI.
- Leukocytospermia >1 million/mL.

I-3f. Principles of antimicrobial therapy in UTI

Type: SAQ (10 marks)

Definitions

- **Uncomplicated** urinary tract infection (UTI): healthy non-pregnant women, normal tract.
Complicated: men, pregnancy, abnormal tract, catheter, immunosuppression.
- **Recurrent:** ≥ 2 in 6 months or ≥ 3 in 12 months (reinfection or persistence).

Principles

- **Site decides the level:** urine in cystitis; tissue in pyelonephritis, prostatitis, bacteraemia. Nitrofurantoin and fosfomycin lack tissue levels: **cystitis only**.
- β -lactams are time-dependent; aminoglycosides and fluoroquinolones (FQs) concentration-dependent (once-daily gentamicin).
- **Resistance thresholds:** trimethoprim-sulfamethoxazole (TMP-SMX) if local resistance $\leq 20\%$; oral FQ for pyelonephritis if $\leq 10\%$.
- **Collateral damage:** FQs and cephalosporins select resistance in bowel flora and *C. difficile*, so avoid them in cystitis.
- Culture all except simple cystitis; **drain** obstructed infected kidneys; shortest effective course.

Empirical choices

Scenario	Drug	Duration
Cystitis (women)	Nitrofurantoin; fosfomycin 3 g; pivmecillinam; TMP-SMX	5 d; single; 3–5 d; 3 d
Cystitis (men)	TMP-SMX or per culture	7 d (= 14 d, Drekonja 2021)
Pyelonephritis (oral)	Ciprofloxacin; levofloxacin 750 mg	7 d; 5 d
Complicated	Aminoglycoside $\pm$ ampicillin, or 3rd-generation cephalosporin	7–14 d
Urosepsis	Broad-spectrum within 1 h , source control	–

Asymptomatic bacteriuria

- Treat **only in pregnancy and before mucosa-breaching urological procedures**.

Prophylaxis

- Recurrent: vaginal oestrogen, **methenamine hippurate** (non-inferior to antibiotics, ALTAR 2022), cranberry, post-coital antibiotics.
- Surgical: single dose within 1 hour; ≤ 24 h for percutaneous nephrolithotomy and prostheses.
- **Also know:** ESBL (extended-spectrum β -lactamase) producers; nitrofurantoin avoided in renal failure; new agents pivmecillinam (US 2024) and gepotidacin (2025).

Diagram to draw: Concentration-time curve showing peak:MIC (minimum inhibitory concentration).

Key points to remember:

- Recurrent = ≥ 2 in 6 months or ≥ 3 in 12 months.
 - Cystitis: nitrofurantoin 5 d, fosfomycin single, TMP-SMX 3 d (resistance $\leq 20\%$).
 - Pyelonephritis: ciprofloxacin 7 d or levofloxacin 5 d (FQ resistance $\leq 10\%$).
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Rapid Revision Sheet - Paper I

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

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- RBF 25% of cardiac output; GFR 125 mL/min; net filtration pressure ~10 mmHg.
- Proximal 65%, loop 25%, distal 5–7% of Na⁺; ADH → V2 → AQP2.
- Myogenic peristalsis from calyceal pacemaker; post-obstructive diuresis 2% vs 50%.

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- DeLancey: I suspension, II attachment (ATFP), III fusion; level I is the keystone.
- Enhorning 1961, DeLancey 1994, Petros & Ulmsten 1990, McGuire ISD 1978.
- Central cystocele rugae lost, paravaginal preserved; POP-Q II = ± 1 cm of hymen.

I-3a. Newer generation CT scan (SAQ)

- Doses: conventional 10–20, low-dose <3–4, ultra-low 1–2 mSv.
- DECT: uric acid stones, virtual unenhanced, iodine maps.
- Enhancement ≥ 20 HU = solid; deep learning enables ~1 mSv CT.

I-3b. History of Laparoscopy (SAQ)

- Bozzini 1806; Kelling 1901 (dog); Jacobaeus 1910 (human); Veress 1938; Hopkins rod-lens.
- Urology: Cortesi 1976; Clayman 1990; Gaur 1992; Ratner 1995.
- da Vinci 2000; Binder first robotic prostatectomy; robot solved suturing barrier.

I-3c. Renin-angiotensin system (SAQ)

- Renin is rate-limiting; released by low pressure, low NaCl, $\beta 1$.
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- Renin ratio >1.5 lateralises; reninoma excision cures most.

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